Relative-fit and nested difference tests for estimated-weight moment-quadratic estimators

June 20, 2026

Purpose

The estimated-weight sandwich (companion note `weighted_moment_ij_grid`) corrects the parameter covariance by adding a weight-derivative term. This note asks whether the same correction extends to model-comparison tests, and whether the nested difference test follows as a corollary. Three answers, stated up front.

1. The correction is proportional to the pseudo-true residual r_* (Proposition 1). It is identically zero for the goodness-of-fit null, where the focal model is assumed correct, and present for every test whose null does not assume correctness.

2. For distinguishable models the relative-fit (Vuong) variance is a clean corollary: $\omega^2 = c^\top \Gamma c$ with the weight channel carried by an explicit term in c (Proposition 2).

3. The nested, degenerate case is *not* a clean corollary. The limiting quadratic form is the Hessian of the profiled discrepancy difference, not the covariance quadratic of the fitted residual fluctuation. The naive substitution $\Gamma \mapsto (W - W_r)\Gamma(W - W_r)^\top$ into the U-matrix is wrong (Proposition 3). It needs its own derivation.

1 Recap of the corrected influence

The estimator minimises $F_N(\theta) = \frac{1}{2} r(\theta)^\top \widehat{W} r(\theta)$ with residual $r(\theta) = \eta(\theta) - u$, Jacobian $\Delta = \partial\eta/\partial\theta^\top$, and weight $\widehat{W} = W(u)$ built from the sample moments u. Write g_i for the moment influence function, $\Gamma = \mathbb{E}[g_i g_i^\top]$, and

θ_*, $r_* = \eta(\theta_*) - u_0$ for the pseudo-true value and residual. The estimator is asymptotically linear with influence function $\phi_i = H^{-1}\Delta^{\mathsf{T}} v_i$, where

$$H = A + C, \quad A = \Delta^{\mathsf{T}} W \Delta, \quad C = \sum_k (W r_*)_k \nabla^2 \eta_k(\theta_*), \quad v_i = W g_i - \dot{W}_i\, r_*, \tag{1}$$

where $\dot{W}_i$ is the casewise influence of the weight. When the weight is a function of the modelled moments u, $\dot{W}_i r_* = W_r g_i$ with $W_r = [\,\partial W/\partial u_1\, r_*,\ \ldots,\ \partial W/\partial u_m\, r_*\,]$ the residual-contracted weight derivative, so $v_i = (W - W_r) g_i$ and the corrected meat is $\Omega = (W - W_r)\Gamma(W - W_r)^{\mathsf{T}}$. In general $\Omega = \mathrm{Var}(v_i)$ is formed from the casewise rows; the matrix form is the special case used in Sections 3–5.

2 The residual governs the correction

Proposition 1. *Every estimated-weight correction in* (1)*, namely the curvature term C and the weight term W_r, is linear in the pseudo-true residual r_*. Hence both vanish exactly when $r_* = 0$. For a test whose null fixes the focal model as correctly specified the correction is identically zero; for a test whose null leaves the focal model misspecified the correction is present.*

Proof. C contracts $W r_*$ and W_r contracts r_* columnwise, so both are $O(\|r_*\|)$ and vanish at $r_* = 0$. The goodness-of-fit null sets $\eta(\theta_*) = u_0$, hence $r_* = 0$. The relative-fit and the not-assuming-correct nested nulls leave $r_* \neq 0$. $\square$

A second, sharper reason closes the goodness-of-fit case. At that null the fitted residual is $r(\widehat{\theta}) = O_p(N^{-1/2})$, so the weight fluctuation enters the residual only through $(\widehat{W} - W) r(\widehat{\theta}) = O_p(N^{-1})$, which is negligible. The estimated weight is therefore asymptotically irrelevant to the fit statistic, which is why the fixed-weight projection theory of [1] and the scaled statistic implemented in practice are already correct under the null.

3 Goodness of fit: the projection, unchanged

For a fixed weight, expand the fitted residual at the goodness-of-fit null. With $H = A$ and $W_r = 0$ there,

$$\sqrt{N}\, r(\widehat{\theta}) = -M\sqrt{N}(u - u_0) + o_p(1), \qquad M = I - \Delta A^{-1} \Delta^{\mathsf{T}} W,$$

so $\mathrm{Var}(\sqrt{N}\, r(\widehat{\theta})) = M\Gamma M^{\mathsf{T}}$ and the fit statistic $N\, r^{\mathsf{T}} W r$ converges to $\sum_j \lambda_j \chi^2_{1,j}$ with $\lambda_j = \mathrm{eig}(U\Gamma)$ and $U = WM = W - W\Delta(\Delta^{\mathsf{T}} W \Delta)^{-1}\Delta^{\mathsf{T}} W$ [1]. When

$W = \Gamma^{-1}$ the product $U\Gamma$ is idempotent and the law collapses to χ^2_{df}; for $W \neq \Gamma^{-1}$ (DWLS, ULS) it is a genuine mixture, the reason the scaled statistic [3] exists. By Proposition 1 the estimated weight contributes nothing here.

4 Relative fit: a clean corollary

Compare two structural models A and B on the same indicators. The weight $\widehat{W} = W(u)$ is a function of the sample moments alone, so it is shared, $W_A = W_B = W$. Let $D = F_A(\widehat{\theta}_A) - F_B(\widehat{\theta}_B)$ be the sample discrepancy difference and r_*^A, r_*^B the two pseudo-true residuals.

Proposition 2. *The discrepancy difference is asymptotically linear,* $\sqrt{N}\, D = N^{-1/2} \sum_i c^{\mathsf{T}} g_i + o_p(1)$*, with*

$$c = -W(r_*^A - r_*^B) + \tfrac{1}{2}(b^A - b^B), \qquad b_l^A = (r_*^A)^{\mathsf{T}} \frac{\partial W}{\partial u_l}\, r_*^A, \tag{2}$$

and b_l^B defined likewise. For distinguishable models $\omega^2 = c^{\mathsf{T}}\Gamma c > 0$ and $\sqrt{N}\, D/\widehat{\omega} \to \mathcal{N}(0,1)$.

Proof. View F_A as a functional of the sampling distribution through u, the minimiser $\widehat{\theta}_A(u)$, and the weight $W(u)$. The minimiser is interior, so the envelope theorem removes the $\widehat{\theta}_A$ channel and

$$\mathrm{IF}_i(F_A) = \Big(\tfrac{\partial F_A}{\partial u} + \tfrac{\partial F_A}{\partial W}\, DW(u_0)\Big) g_i = \big(-W r_*^A + \tfrac{1}{2} b^A\big)^{\mathsf{T}} g_i,$$

using $\partial F_A/\partial u = -r^{\mathsf{T}} W$ and $\partial(\frac{1}{2} r^{\mathsf{T}} W r)/\partial W$ contracted with $DW[g_i] = \sum_l (\partial W/\partial u_l) g_{i,l}$. Subtracting $\mathrm{IF}_i(F_B)$ gives (2); the central limit theorem gives the normal law. □

The term $-W(r_*^A - r_*^B)$ is the difference of weighted residuals that a fixed-weight account would give. The term $\frac{1}{2}(b^A - b^B)$ is the weight channel, quadratic in each model's own residual and contracted with the weight derivative. It is the testing analogue of W_r and is the new ingredient. Every quantity is already available: r_*^A, r_*^B from the two fits, $\partial W/\partial u$ from the same derivative the standard error uses, and Γ from the stage-one moments. The relative-fit test is therefore a corollary of the standard-error machinery, with no projection involved because the envelope theorem has removed the parameter channel.

5 The nested case does not commute

When B is nested in A the discrepancy difference degenerates under the null that the restriction holds, $r_*^A = r_*^B$ in the relevant directions, so $c = 0$ and $\omega^2 = 0$. The first-order term vanishes and the limiting law is the second-order quadratic form. The question is what second derivative that quadratic form uses.

For one model M, let

$$f_M(u) = \min_\theta \tfrac{1}{2} r_M(\theta, u)^\mathsf{T} W(u) r_M(\theta, u), \qquad r_M(\theta, u) = \eta_M(\theta) - u.$$

The envelope gradient of this profiled scalar discrepancy is

$$s_M(u) = \nabla_u f_M(u) = -W r_*^M + \tfrac{1}{2} b_M, \qquad (b_M)_l = (r_*^M)^\mathsf{T} \frac{\partial W}{\partial u_l} r_*^M. \tag{3}$$

This is the "score" used below: a moment-space / profile-value score, not the parameter score $\partial F/\partial\theta$.

Proposition 3. *Suppose the nested null is true at the pseudo-true level: the larger model's pseudo-true point lies in the smaller model, so the two profiled values and profile scores agree at u_0, $f_A(u_0) = f_B(u_0)$ and $s_A(u_0) = s_B(u_0)$. Let*

$$Q_M = D_u s_M(u_0)$$

be the Hessian of the profiled discrepancy for model M. Then

$$2N\{f_B(\widehat{u}) - f_A(\widehat{u})\} \xrightarrow{d} z^\mathsf{T}(Q_B - Q_A) z, \qquad z \sim N(0, \Gamma),$$

and the reference spectrum is

$$\lambda_j = \operatorname{eig}\{(Q_B - Q_A)\Gamma\}.$$

For a fixed weight,

$$Q_M = W - W\Delta_M H_M^{-1} \Delta_M^\mathsf{T} W, \qquad H_M = A_M + C_M. \tag{4}$$

For an estimated weight $W(u)$, Q_M is the derivative of (3). It contains the fitted-residual derivative

$$D_u r_*^M[h] = \Delta_M H_M^{-1} \Delta_M^\mathsf{T} \{Wh - DW[h] r_*^M\} - h$$

and the first and second derivatives of the weight map. Consequently the nested law is not obtained by replacing Γ by $(W - W_r)\Gamma(W - W_r)^\mathsf{T}$ in Satorra's fixed-weight U matrix.

Proof. Taylor expand the two profiled scalar functions around u_0:

$$f_M(u_0 + N^{-1/2}z) = f_M(u_0) + N^{-1/2}s_M(u_0)^\mathsf{T} z + \tfrac{1}{2}N^{-1}z^\mathsf{T} Q_M z + o_p(N^{-1}).$$

The nested pseudo-null cancels the value and first derivative. Multiplying the difference by $2N$ leaves $z^\mathsf{T}(Q_B - Q_A)z + o_p(1)$, giving the eigenvalue law. For fixed W, differentiating the population first-order condition $\Delta^\mathsf{T} W r_* = 0$ gives $D\theta[h] = H^{-1}\Delta^\mathsf{T} W h$ and $Dr_*[h] = \Delta H^{-1}\Delta^\mathsf{T} W h - h$. Since $s = -Wr_*$ when W is fixed, $Q[h] = -WDr_*[h]$, which is (4). When $W = W(u)$, the same calculation includes $DW[h]r_*$ in the first-order condition and the derivative of the weight term in (3); this introduces the first and second weight derivatives. □

The fitted residual fluctuation is still useful for standard errors:

$$\sqrt{N}\,(\hat{r}_M - r_*^M) = -\widetilde{M}_M z + o_p(1), \qquad \widetilde{M}_M = I - \Delta_M H_M^{-1}\Delta_M^\mathsf{T}(W - W_r^M)$$

when $W = W(u)$. But the covariance quadratic $\widetilde{M}_M^\mathsf{T} W \widetilde{M}_M$ is not the Hessian of the profiled discrepancy unless $H_M = A_M$ and the weight terms satisfy special cancellations. The nested test is a profile-value problem, not a residual-vcov problem.

6 Specialization: complete-data DWLS on a linear covariance model

The curvature term $C = \sum_k (Wr_*)_k \nabla^2 \eta_k(\theta_*)$ is the only place the shape of the model surface enters. It vanishes whenever η is linear in θ, which removes one of the two deformations and isolates the weight channel.

6.1 A model with $C = 0$

Take a linear covariance structure $\Sigma(\theta) = \sum_k \theta_k G_k$ with fixed symmetric G_k, so $\eta(\theta) = \operatorname{vech}\Sigma(\theta) = \Delta\theta$ with $\Delta = [\operatorname{vech} G_1, \ldots, \operatorname{vech} G_q]$ constant, $\nabla^2\eta_k = 0$, hence $C = 0$ and $H = A = \Delta^\mathsf{T} W \Delta$ constant. A concrete nested pair on p continuous variables: B is compound symmetry (one variance, one covariance), nested in A (one variance, free covariances), the difference testing equality of the covariances. Both are linear, so the only surviving deformation is the weight channel.

Remark (A test bed, not the headline model). The parameter is restricted to the open cone $\{\theta : \Sigma(\theta) \succ 0\}$; for compound symmetry this is the wedge

$a - b > 0$ and $a + (p-1)b > 0$ in $\theta = (a, b) = (\text{variance}, \text{covariance})$. This is an inequality constraint, inactive at an interior pseudo-true value, so it does not enter the first-order conditions or the asymptotics, and $C = 0$ holds from the linearity of η regardless of it. The only design requirement is to keep the data-generating θ_* interior, with no near-singular best fit. Linear covariance structures are a classical class (variance components, exchangeable and Toeplitz patterns, fixed-loading factor models), but pairing one with DWLS is off the beaten path. Its role here is a curvature-free check: a deformation that appears here cannot be curvature in disguise. The substantive case keeps a genuine factor model and carries $C \neq 0$ (Section 6.6).

6.2 DWLS ingredients

The stage-one moments are $u = \text{vech}(S)$ with casewise influence $g_i = \text{vech}(z_i z_i^\mathsf{T}) - \sigma_0$, $z_i = y_i - \mu$, and $\Gamma = \mathbb{E}[g_i g_i^\mathsf{T}]$ the Browne fourth-moment matrix. DWLS uses $\widehat{W} = \text{diag}(\widehat{\Gamma})^{-1}$, so with $\gamma = \text{diag}(\Gamma)$ the population weight is $W = \text{diag}(\gamma)^{-1}$. The weight influence acts diagonally,

$$\dot{w}_{i,k} = -\gamma_k^{-2}\,\dot{\gamma}_{i,k}, \qquad \dot{\gamma}_{i,k} = g_{i,k}^2 - \gamma_k + (\text{centring corrections}),$$

where $\dot{\gamma}_{i,k}$ is the casewise influence of the k-th diagonal of $\widehat{\Gamma}$. The corrected casewise moment-space influence is

$$v_{i,k} = \frac{g_{i,k}}{\gamma_k} + \frac{r_{*,k}\,\dot{\gamma}_{i,k}}{\gamma_k^2}, \tag{5}$$

which matches the implemented DWLS adapter.

Remark (The DWLS weight channel is not a matrix on g_i). $\widehat{W} = \text{diag}(\widehat{\Gamma})^{-1}$ depends on fourth-order sample moments, not on the modelled $u = \text{vech}(S)$. Its influence $\dot{\gamma}_{i,k}$ is quadratic in g_i (through $g_{i,k}^2$), so the weight channel in (5) is a genuine casewise quantity, not $W_r g_i$ for a matrix W_r. The matrix annihilator $\widetilde{M}$ of Proposition 3 therefore holds only when the weight is a function of u; for DWLS the residual influence $\rho_i = \Delta A^{-1}\Delta^\mathsf{T} v_i - g_i$ stays casewise and carries a part quadratic in g_i, so its covariance reaches eighth-order moments of the data.

6.3 The verification-friendly twin: normal-theory weight

The normal-theory weight $W = W_{\text{NT}}(S)$ is a function of u, so it is the clean case. With $C = 0$, the residual derivative uses $\widetilde{M} = I - \Delta A^{-1}\Delta^\mathsf{T}(W - W_r)$, but the nested test uses $Q = D_u s$, not $\widetilde{M}^\mathsf{T} W \widetilde{M}$. A fixed weight would give $Q = U = W - W\Delta A^{-1}\Delta^{\mathsf{T}}W$, and with $W = \Gamma^{-1}$ the usual Satorra spectrum collapses to χ^2_{df}. A data-estimated normal-theory weight adds the derivative of the envelope score (3), including second derivatives of $W_{\mathrm{NT}}(u)$. This isolates the weight channel from curvature while keeping the actual test statistic in view.

6.4 The geometry of the payoff

The deformation of the profile Hessian is zero at $r_* = 0$ and analytic in r_*. The exploration is to sweep the direction of r_* at fixed $\|r_*\|$: generate from $\Sigma_0 = \Sigma_B(\theta_*) + \epsilon D$ and rotate D from W-orthogonal to the constraint space toward aligned. The conjecture, which the simulation tests rather than assumes, is that orthogonal r_* leaves the law near Satorra (the benign regime where robust intervals already calibrate) and aligned r_* deforms it most. The script `relative-and-nested-fit-check.py` computes $Q_B - Q_A$ by finite differences of the profile score and compares it with the ordinary Satorra spectrum and with the residual-fluctuation quadratic.

6.5 The residual deformation in closed form

With $C = 0$ the fitted-residual fluctuation has an exact deformation. This is a useful diagnostic for the standard-error machinery, but it is not the nested test law once the weight is data-estimated.

Proposition 4 (Closed-form deformation under $C = 0$)**.** *Let $H = A = \Delta^{\mathsf{T}}W\Delta$ and $\widetilde{M} = I - \Delta A^{-1}\Delta^{\mathsf{T}}(W - W_r)$. Then*

$$\widetilde{M}^{\mathsf{T}}W\widetilde{M} = U + R, \qquad U = M^{\mathsf{T}}WM, \qquad R = (\Delta^{\mathsf{T}}W_r)^{\mathsf{T}}A^{-1}(\Delta^{\mathsf{T}}W_r) \succeq 0,$$

the cross terms vanishing by the annihilator orthogonality $\Delta^{\mathsf{T}}WM = 0$. Hence the residual-vcov quadratic has spectrum $\mathrm{eig}[(U_B - U_A + R_B - R_A)\Gamma]$. *This is not, in general, the spectrum of $2N(f_B - f_A)$; the latter is* $\mathrm{eig}[(Q_B - Q_A)\Gamma]$ *from Proposition 3.*

Proof. Write $\widetilde{M} = M + \delta$ with $\delta = \Delta A^{-1}\Delta^{\mathsf{T}}W_r \in \mathrm{col}(\Delta)$. Since $M^{\mathsf{T}}W\Delta = W\Delta - W\Delta A^{-1}(\Delta^{\mathsf{T}}W\Delta) = 0$, both cross terms $M^{\mathsf{T}}W\delta$ and $\delta^{\mathsf{T}}WM$ vanish, and $\delta^{\mathsf{T}}W\delta = W_r^{\mathsf{T}}\Delta A^{-1}(\Delta^{\mathsf{T}}W\Delta)A^{-1}\Delta^{\mathsf{T}}W_r = (\Delta^{\mathsf{T}}W_r)^{\mathsf{T}}A^{-1}(\Delta^{\mathsf{T}}W_r) = R$. □

The consequence is narrower than first hoped: W_r is linear in r_*, so this residual-vcov deformation is quadratic in r_* and vanishes when a model fits. That statement validates the residual influence algebra, not the second-order law of the profiled discrepancy. The numerical check confirms the identity to machine precision and then separately computes the profile-Hessian law.

Remark (Why this is special to $C = 0$). The cancellation uses $A = \Delta^\mathsf{T} W \Delta$. With curvature the bread is $H = A + C$, $M^\mathsf{T} W\Delta = W\Delta - W\Delta H^{-1} A \neq 0$, and the cross terms survive. The clean $U + R$ split is exactly what the entanglement of Proposition 3 breaks; the linear test bed buys a closed form that the factor model does not have.

6.6 The substantive twin: a factor model with $C \neq 0$

Dropping linearity costs the curvature term. For a confirmatory factor model $\Sigma(\theta) = \Lambda\Psi\Lambda^\mathsf{T} + \Theta$ the moments are quadratic in the loadings, so $\nabla^2 \eta_k$ is sparse and closed form (the $\lambda_a\lambda_b$ blocks), and $H = A + C$ with $C = \sum_k (W r_*)_k \nabla^2 \eta_k(\theta_*)$ evaluated at the pseudo-true point. For fixed weights the profile Hessian is $Q = W - W\Delta H^{-1}\Delta^\mathsf{T} W$, so replacing A by H is already necessary. For estimated weights, the derivative of (3) adds the weight channel on top. A natural nested pair is configural versus metric invariance across two groups, the difference testing equality of a block of loadings. The price of realism is that the deformation now superposes two channels, curvature and weight, so the clean attribution of the linear test bed is lost. The CFA check script verifies that the old residual-vcov quadratic breaks away from the finite-difference profile-Hessian law when $C \neq 0$.

7 Verdict

The testing extension partitions cleanly. The goodness-of-fit test needs no correction (Proposition 1); the prevailing scaled statistic is already right. The relative-fit test for distinguishable models is a corollary of the standard-error machinery, with $\omega^2 = c^\mathsf{T}\Gamma c$ and the weight channel (2) computable from quantities the standard error already forms (Proposition 2). The nested difference test, in the regime that does not assume the larger model correct, is its own project, because the object is the Hessian of the profiled discrepancy difference. The fitted residual influence and the parameter sandwich supply ingredients, but the profile-Hessian law also needs curvature and, for estimated weights, second derivatives of the weight map (Proposition 3). For the manuscript this argues to keep the estimated-weight standard error and the distinguishable relative-fit test together, and to treat the nested weighted-χ^2 law as a sequel [4, 5].

References

[1] Satorra, A. (1989). Alternative test criteria in covariance structure analysis: a unified approach. *Psychometrika*, 54(1), 131–151. doi:10.1007/BF02294453.

[2] Satorra, A. (2000). Scaled and adjusted restricted tests in multi-sample analysis of moment structures. In R. D. H. Heijmans, D. S. G. Pollock, & A. Satorra (Eds.), *Innovations in Multivariate Statistical Analysis* (pp. 233–247). Springer. doi:10.1007/978-1-4615-4603-0_17.

[3] Satorra, A., & Bentler, P. M. (2001). A scaled difference chi-square test statistic for moment structure analysis. *Psychometrika*, 66(4), 507–514. doi:10.1007/BF02296192.

[4] Vuong, Q. H. (1989). Likelihood ratio tests for model selection and nonnested hypotheses. *Econometrica*, 57(2), 307–333. doi:10.2307/1912557.

[5] Merkle, E. C., You, D., & Preacher, K. J. (2016). Testing nonnested structural equation models. *Psychological Methods*, 21(2), 151–163. doi:10.1037/met0000038.

[6] Lai, K., & Simões, F. (2023). Reflecting on the "robust" standard errors for two-stage structural equation modeling. *Structural Equation Modeling*, 30(5).

[7] Hall, A. R., & Inoue, A. (2003). The large sample behaviour of the generalized method of moments estimator in misspecified models. *Journal of Econometrics*, 114(2), 361–394. doi:10.1016/S0304-4076(03)00075-0.